

RUNZE SHI

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EDUCATION

University of British Columbia

Vancouver, BC

Master of Applied Science - Mechanical Engineering, GPA: 4.10/4.33

Sep. 2024 – Present

University of Wisconsin-Madison

Madison, WI

B.S. Mechanical Engineering, GPA: 3.75/4.0 (Top 10%)

Sep. 2020 – May. 2024

Core Courses: FEA, Computational Solid Mechanics, Heat Transfer, Mechatronics, System Dynamics and Control

TECHNICAL SKILLS

Simulation: ANSYS Workbench/APDL, Abaqus, Femap, COMSOL, static, modal, harmonic, buckling

Mechanical Design & Manufacturing: SolidWorks, AutoCAD, GD&T fundamentals, mechanical drawings

Programming: Python, MATLAB/Simulink, Fortran, Git/GitHub (version control)

EXPERIENCE

Laboratory for Structural Efficiency (LASE), UBC | *Graduate Research Assistant*

Sep. 2024 – Present

- Developed and validated nonlinear coupled thermal-mechanical Abaqus models for welded structures using temperature-dependent materials, element activation, and a custom Fortran DFLUX heat-source subroutine.
- Automated Abaqus simulation setup, result extraction and Python-based post-processing workflows to generate structured FEA datasets, enabling faster design iterations, and cross-comparison of parameters and configurations.
- Built ML surrogate models to predict element-level inherent strain from FEA data, reducing reliance on full thermal-mechanical simulations and accelerating welding distortion prediction by over 90%.

Mixed-Gas Joule-Thomson Cryocooling System, SEL Lab | *Research Assistant*

Mar. 2023 – May. 2024

- Redesigned the tubing layout of a mixed-gas compressor station and integrated additional gas tanks, pressure regulators, and flowmeters using brazed VCR and NPT connections.
- Restored a discontinued 20-year-old data-acquisition system by reverse-engineering its operating protocols, configuring 16 sensors, and developing LabVIEW pipelines for pressure, temperature, and flow monitoring.
- Diagnosed leakage and instrumentation issues in the high-pressure loop through sensor-data analysis and system-level troubleshooting, improving operating safety, data reliability, and test readiness.

State Laboratory of Fluid Power and Mechatronic Systems | *Robotics Integration Intern*

Jun. 2023 – Aug. 2023

- Revised SolidWorks assemblies and detailed engineering drawings for a 4-DOF underwater manipulator, adding a wrist-gripper rotation mechanism while maintaining servo interfaces and assembly constraints.
- Supported mechanical integration of the manipulator with an autonomous underwater vehicle and assisted with system setup, troubleshooting, and underwater testing at pressures up to 1.5 atm.
- Developed MATLAB/Simulink kinematic models to support motion analysis, control tuning, and comparison between simulated and experimental system behavior.

DJI (Da-Jiang Innovations) | *Data & Intelligent Algorithms Intern*

Jun. 2025 – Sep. 2025

- Developed automated Python pipelines to clean, validate, and process 18,000+ perception samples, improving data quality and reducing manual preprocessing for model development.
- Investigated model-performance issues across different operating conditions and implemented preprocessing and parameter improvements that increased classification accuracy from 82% to 97%.

SELECTED ENGINEERING PROJECTS

Stiffened Panel Buckling Optimization

Mar. 2025 – Apr. 2025

- Developed and parametrically evaluated Femap models of stiffened steel panels under uniaxial compression, varying plate and stiffener geometry under a fixed material-area constraint.
- Performed linear eigenvalue and nonlinear buckling analyses to estimate the critical load, identify the first buckling mode, and characterize the load-displacement response near instability.

Passive Thermal Design of a 6U Small Satellite

May. 2026 – Jun. 2026

- Developed a transient thermal-network model for a 6U satellite in a 600 km sun-synchronous orbit and designed a passive thermal-control configuration using surface zoning, optical coatings, body-mounted solar cells, and a radiator patch.